Patient Information	Specimen Information	Client Information
NEUGEBAUER, ANTON **DOB: 12/12/1992 AGE: 28** Gender: M Fasting: Y Phone: 610.357.4409 Patient ID: 12121992NA Health ID: 8573027291189697	Specimen: MR775705U Requisition: 0000004 Lab Ref #: 15093263 Collected: 05/18/2021 / 09:00 EDT Received: 05/19/2021 / 05:07 EDT Faxed: 06/16/2021 / 12:23 EDT	Client #: 73912401 MAIL992 REPIK, MICHAEL R P2P AGE MANAGEMENT 133 E DE LA GUERRA ST # 425 SANTA BARBARA, CA 93101-2228

COMMENTS: FASTING:YES

Test Name	In Range	Out Of Range	Reference Range	Lab
IRON, TIBC AND FERRITIN PANEL				
IRON AND TOTAL IRON BINDING CAPACITY				MI
IRON, TOTAL	62		50-195 mcg/dL	
IRON BINDING CAPACITY	368		250-425 mcg/dL (calc)	
% SATURATION		**17 L**	20-48 % (calc)	
FERRITIN	76		38-380 ng/mL	MI
THYROID PANEL WITH TSH				
THYROID PANEL				MI
T3 UPTAKE	33		22-35 %	
T4 (THYROXINE), TOTAL	7.6		4.9-10.5 mcg/dL	
FREE T4 INDEX (T7)	2.5		1.4-3.8	
TSH	1.26		0.40-4.50 mIU/L	MI
LIPID PANEL, STANDARD				
CHOLESTEROL, TOTAL	129		<200 mg/dL	MI
HDL CHOLESTEROL	41		> OR = 40 mg/dL	MI
TRIGLYCERIDES	68		<150 mg/dL	MI
LDL-CHOLESTEROL	74		mg/dL (calc)	MI
Reference range: <100 Desirable range <100 mg/dL for primary prevention; <70 mg/dL for patients with CHD or diabetic patients with > or = 2 CHD risk factors. LDL-C is now calculated using the Martin-Hopkins calculation, which is a validated novel method providing better accuracy than the Friedewald equation in the estimation of LDL-C. Martin SS et al. JAMA. 2013;310(19): 2061-2068 (http://education.QuestDiagnostics.com/faq/FAQ164)				
CHOL/HDLC RATIO	3.1		<5.0 (calc)	MI
NON HDL CHOLESTEROL	88		<130 mg/dL (calc)	MI
For patients with diabetes plus 1 major ASCVD risk factor, treating to a non-HDL-C goal of <100 mg/dL (LDL-C of <70 mg/dL) is considered a therapeutic option.				
HOMOCYSTEINE	7.3		<11.4 umol/L	MI
Homocysteine is increased by functional deficiency of folate or vitamin B12. Testing for methylmalonic acid differentiates between these deficiencies. Other causes of increased homocysteine include renal failure, folate antagonists such as methotrexate and phenytoin, and exposure to nitrous oxide. Selhub J, et al., Ann Intern Med. 1999;131(5):331-9.				
COMPREHENSIVE METABOLIC PANEL				MI
GLUCOSE	91		65-99 mg/dL Fasting reference interval	
UREA NITROGEN (BUN)	12		7-25 mg/dL	
CREATININE	0.82		0.60-1.35 mg/dL	

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Test Name	In Range	Out Of Range	Reference Range	Lab
eGFR NON-AFR. AMERICAN	120		> OR = 60 mL/min/1.73m2	
eGFR AFRICAN AMERICAN	139		> OR = 60 mL/min/1.73m2	
BUN/CREATININE RATIO	NOT APPLICABLE		6-22 (calc)	
SODIUM	140		135-146 mmol/L	
POTASSIUM	4.8		3.5-5.3 mmol/L	
CHLORIDE	103		98-110 mmol/L	
CARBON DIOXIDE	29		20-32 mmol/L	
CALCIUM	9.8		8.6-10.3 mg/dL	
PROTEIN, TOTAL	7.3		6.1-8.1 g/dL	
ALBUMIN	4.7		3.6-5.1 g/dL	
GLOBULIN	2.6		1.9-3.7 g/dL (calc)	
ALBUMIN/GLOBULIN RATIO	1.8		1.0-2.5 (calc)	
BILIRUBIN, TOTAL		**1.5 H**	0.2-1.2 mg/dL	
ALKALINE PHOSPHATASE	86		36-130 U/L	
AST	17		10-40 U/L	
ALT	11		9-46 U/L	
HEMOGLOBIN A1c	4.9		<5.7 % of total Hgb	MI

For the purpose of screening for the presence of diabetes:

<5.7%	Consistent with the absence of diabetes
5.7-6.4%	Consistent with increased risk for diabetes (prediabetes)
> or =6.5%	Consistent with diabetes

This assay result is consistent with a decreased risk of diabetes.

Currently, no consensus exists regarding use of hemoglobin A1c for diagnosis of diabetes in children.

According to American Diabetes Association (ADA) guidelines, hemoglobin A1c <7.0% represents optimal control in non-pregnant diabetic patients. Different metrics may apply to specific patient populations. Standards of Medical Care in Diabetes(ADA).

Test Name	In Range	Out Of Range	Reference Range	Lab
T3, FREE	3.8		2.3-4.2 pg/mL	MI
THYROID PEROXIDASE AND THYROGLOBULIN ANTIBODIES				
THYROGLOBULIN ANTIBODIES	<1		< or = 1 IU/mL	MI
THYROID PEROXIDASE ANTIBODIES	<1		<9 IU/mL	MI
IGF 1, LC/MS	177		63-373 ng/mL	EZ
Z SCORE (MALE)	0.2		-2.0 - +2.0 SD	

This test was developed and its analytical performance characteristics have been determined by Quest Diagnostics Nichols Institute San Juan Capistrano. It has not been cleared or approved by FDA. This assay has been validated pursuant to the CLIA regulations and is used for clinical purposes.

Test Name	In Range	Out Of Range	Reference Range	Lab
CBC (INCLUDES DIFF/PLT)				MI
WHITE BLOOD CELL COUNT	5.3		3.8-10.8 Thousand/uL	
RED BLOOD CELL COUNT	5.16		4.20-5.80 Million/uL	
HEMOGLOBIN	15.2		13.2-17.1 g/dL	

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Test Name	In Range	Out Of Range	Reference Range	Lab
HEMATOCRIT	45.8		38.5-50.0 %	
MCV	88.8		80.0-100.0 fL	
MCH	29.5		27.0-33.0 pg	
MCHC	33.2		32.0-36.0 g/dL	
RDW	12.1		11.0-15.0 %	
PLATELET COUNT	272		140-400 Thousand/uL	
MPV	10.0		7.5-12.5 fL	
ABSOLUTE NEUTROPHILS	2820		1500-7800 cells/uL	
ABSOLUTE LYMPHOCYTES	1839		850-3900 cells/uL	
ABSOLUTE MONOCYTES	498		200-950 cells/uL	
ABSOLUTE EOSINOPHILS	101		15-500 cells/uL	
ABSOLUTE BASOPHILS	42		0-200 cells/uL	
NEUTROPHILS	53.2		%	
LYMPHOCYTES	34.7		%	
MONOCYTES	9.4		%	
EOSINOPHILS	1.9		%	
BASOPHILS	0.8		%	
VITAMIN B12	561		200-1100 pg/mL	MI
DHEA SULFATE	331		85-690 mcg/dL	MI
FSH	3.2		1.6-8.0 mIU/mL	MI
LH	3.1		1.5-9.3 mIU/mL	MI
ESTRADIOL	21		< OR = 39 pg/mL	MI

Reference range established on post-pubertal patient population. No pre-pubertal reference range established using this assay. For any patients for whom low Estradiol levels are anticipated (e.g. males, pre-pubertal children and hypogonadal/post-menopausal females), the Quest Diagnostics Nichols Institute Estradiol, Ultrasensitive, LCMSMS assay is recommended (order code 30289).

Please note: patients being treated with the drug fulvestrant (Faslodex(R)) have demonstrated significant interference in immunoassay methods for estradiol measurement. The cross reactivity could lead to falsely elevated estradiol test results leading to an inappropriate clinical assessment of estrogen status. Quest Diagnostics order code 30289-Estradiol, Ultrasensitive LC/MS/MS demonstrates negligible cross reactivity with fulvestrant.

Test Name	In Range	Out Of Range	Reference Range	Lab
CORTISOL, A.M.	8.7		mcg/dL	MI

Reference Range
8 a.m. (7-9 a.m.) Specimen: 4.0-22.0

Test Name	In Range	Out Of Range	Reference Range	Lab
PSA, TOTAL	0.7		< OR = 4.0 ng/mL	MI

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Test Name	In Range	Out Of Range	Reference Range	Lab
value, should not be interpreted as absolute evidence of the presence or absence of disease.				
TESTOSTERONE, FREE (DIALYSIS) AND TOTAL,MS				AMD
TESTOSTERONE, TOTAL, MS	624		250-1100 ng/dL	

For additional information, please refer to
http://education.questdiagnostics.com/faq/
TotalTestosteroneLCMSMSFAQ165
(This link is being provided for informational/
educational purposes only.)

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Test Name	In Range	Out Of Range	Reference Range	Lab
TESTOSTERONE, FREE	102.3		35.0-155.0 pg/mL	

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Endocrinology

Test Name	Result	Reference Range	Lab
VITAMIN D,25-OH,TOTAL,IA	38	30-100 ng/mL	MI

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Vitamin D Status          25-OH Vitamin D:

Deficiency:                    <20 ng/mL
Insufficiency:              20 - 29 ng/mL
Optimal:                  > or = 30 ng/mL
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Cardio IQ®

Test Name	Current Result & Risk: Optimal	Current Result & Risk: Non-Optimal	Risk/Reference Interval: Optimal	Risk/Reference Interval: Moderate	Risk/Reference Interval: High	Units	Historical Result & Risk
INFLAMMATION							
HS CRP	<0.3		<1.0	1.0-3.0	>3.0	mg/L	

For details on reference ranges please refer to the reference range/comment section of the report.

4myheart Diet & Exercise Coaching Program: Need help achieving and maintaining an optimal weight? Managing stress? Trying to improve physical fitness levels? The 4myheart program provides support and personalized lifestyle guidance to help improve heart health. Please talk to your provider, visit 4myheart.com or call 1-800-432-7889 opt 2 to learn more.

Medical Information For Healthcare Providers: If you have questions about any of the tests in our Cardio IQ offering, please call Client Services at our Quest Diagnostics-Cleveland HeartLab Cardiometabolic Center of Excellence. They can be reached at 866.358.9828, option 1 to arrange a consult with our clinical education team.

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Reference Range/Comments

Analyte Name	In Range	Out Range	Reference Range	Lab
HS CRP	**<0.3**		<1.0 mg/L	Z4M

The AHA/CDC Guidelines recommend hs-CRP ranges for identifying Relative Cardiovascular Risk in patients ages >17 years: <1.0 mg/L Lower Relative Cardiovascular Risk; 1.0-3.0 mg/L Average Relative Cardiovascular Risk; 3.1-10.0 mg/L Higher Relative Cardiovascular Risk. For patients with higher cardiovascular risk, consider retesting in 1-2 weeks to exclude a benign transient elevation secondary to infection or inflammation from the baseline CRP value. Persistent elevations of >10.0 mg/L upon retesting may be associated with infection and inflammation. The AHA/CDC recommendations are based on Pearson TA et al. Circulation. 2003;107:499-511.

PERFORMING SITE:

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